DO NOW — RECALL

Network types, architecture and hardware

Name: _____ Class: _____ Date: _____

1. What does an OS device driver do?

2. What does a router connect?

3. Define a client-server relationship.

4. Binary search on sorted indexes 0–14 for 13: show midpoint index sequence.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Offers an interface to hardware.

2. Model answer

Different IP networks.

3. Model answer

Client requests service from a dedicated server.

4. Model answer

mid7, mid11, mid13; target found at index13.

Vocabulary — know these terms

LAN — Local Area Network.

WAN — Wide Area Network.

switch — Forwards frames inside local network.

router — Forwards packets between IP networks.

access point — Provides wireless connection to LAN.

NEW LEARNING

Read, annotate and explain

Understand

A LAN covers a local area such as a school building; a WAN connects networks across wider geography.

Trace

Client-server uses dedicated service systems; peer-to-peer allows peers to exchange roles and resources, with different management implications.

Explain

NICs connect devices, switches forward local Ethernet frames, wireless APs bridge Wi-Fi, and routers forward IP packets between networks.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace school laptop reaching an external site through a wired school network.

Predict

Switch moves frames within LAN

Check the answer

router handles packets destined for another network.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define LAN, WAN, switch and router. [4]

Question 2

Describe a laptop's network path via wireless access point, switch and router to the Internet. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare client-server and peer-to-peer for school file sharing, giving strengths and limits. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A network switch and router are always interchangeable because they both move packets.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.